

Introduction

Inspired by TV Series *Schitt's Creek*, I wonder what kind of analysis could help Wendy Kurtz, owner of Blouse Barn—a clothing retailer shop—if her new hire, David Rose, were to use his Power BI skills to help Wendy make better business decisions (unlike what she did in the series).

I used a shopping behavior dataset and imagined it to be Blouse Barn's data. I will be taking the role of David to provide key insights and recommendations to improve Blouse Barn's profitability and repeat customer purchases.

Key Insights:

- The data shows that the core customer base of Blouse Barn are aged 20-70 years old with a demographic's majority consisting of *Men*.
- Customers aged 20-30 gave the highest ratings at 3.79 (1.07% above average). Conversely, despite being the highest spenders, the 50-60 age group gave the lowest ratings at 3.71 (1.07% below average).
- *Clothing* generated the most profit followed by *Accessories*. However, there is a noticeable difference between their review ratings with *Accessories* being received better by customers.
- Seasons are seen having prominent changes in review ratings and purchased items.
- Strategies in highlighting visibility in marketing campaigns and product bundling with popular products might help in increasing the volume of sales for *Accessories* while using additional profit to research and improve *Clothing's* quality to resolve its challenges.
- Implementing targeted feedback incentives is recommended to understand preferences better. executing targeted marketing strategies designed for *Women (i.e discounts during Women's Month)* is recommended to drive a stronger customer base in Women's market.
- Blouse Barn's inventory management should be able to adjust to size preferences (with Medium as dominant choice) and different seasons (items that fit the season—Jackets during Fall and Sunglasses during Summer) to accommodate demands and seasonal changes.

Data

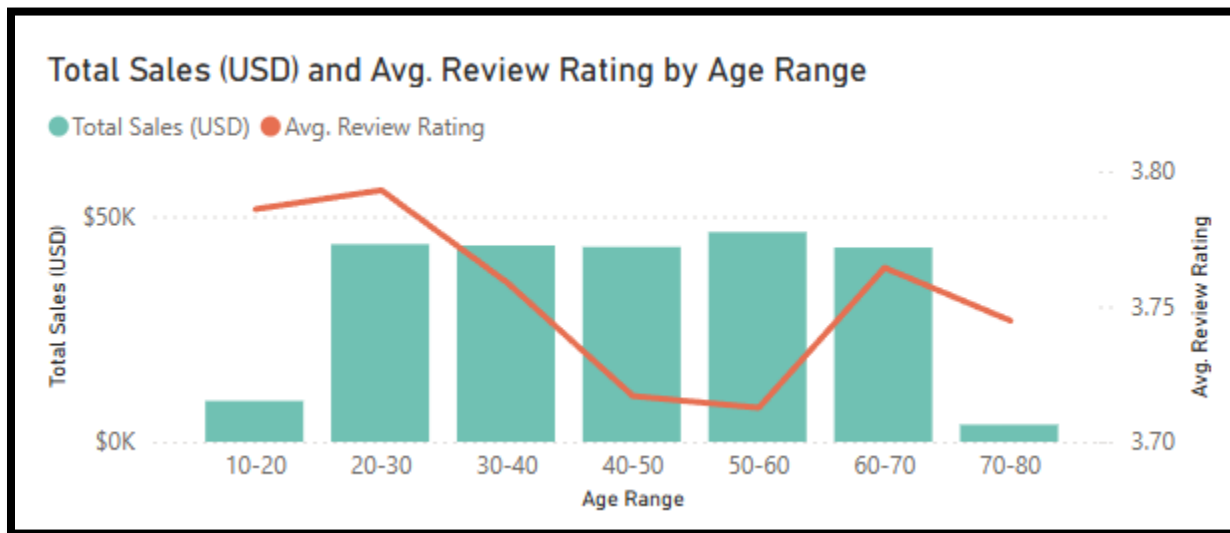
The dataset came from Kaggle with 3900 rows and 18 columns. Checks are done to ensure data's uniqueness and consistency for each variable. There are no missing values and are reviewed and confirmed to fall within expected ranges with no extreme outliers. Here is the link of the notebook for full data preparation: https://colab.research.google.com/drive/1qVHNV-fCCNL4goii8npFpv3U6big_ZpN?usp=sharing

The Analysis

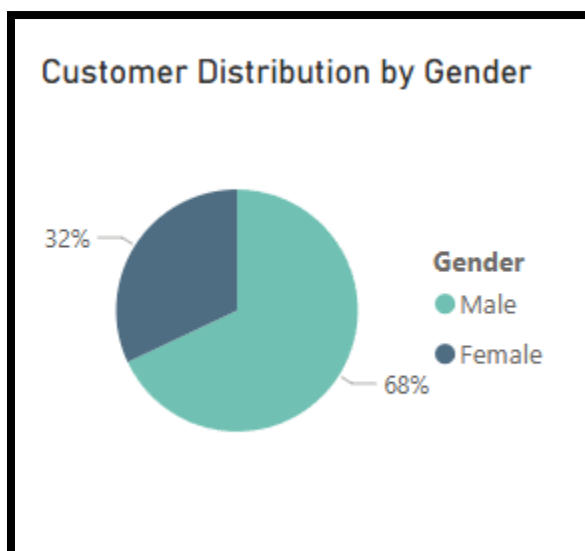
Total Sales and Review Rating by Age Groups:

Blouse Barn generated \$233.08K in total sales, with an average customer spend of \$59.76 per item. An analysis of customer age distribution reveals that while the core customer base spans ages 20 to 70, the 50-60 age group is the highest-spending demographic.

To evaluate customer satisfaction across these demographics, we analyzed average review ratings. While the overall average rating is 3.75 out of 5, satisfaction varies by age. Customers aged 20-30 gave the highest ratings at 3.79 (1.07% above average). Conversely, despite being the highest spenders, the 50-60 age group gave the lowest ratings at 3.71 (1.07% below average).



Blouse Barn team should review feedback from customers aged 40-60 years old to understand why they report lower satisfaction. Implementing a targeted feedback incentive (e.g., a discount on their next purchase) will encourage this group to share specific product insights, allowing the team to better accommodate to their preferences.

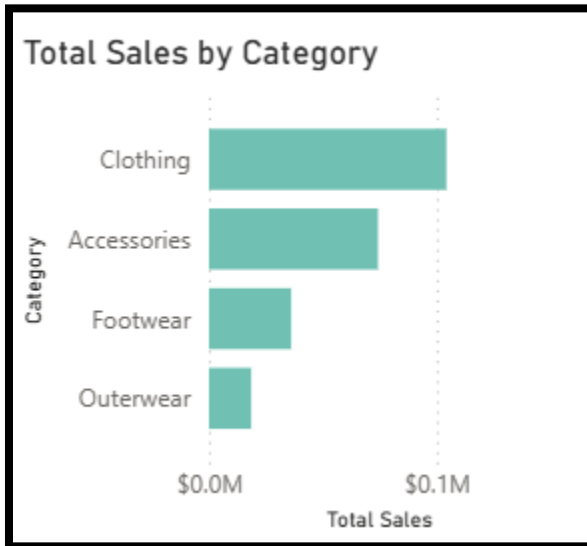


Customer Demographics:

There are 3900 customers in total that Blouse Barn received throughout the year. The greater part of it is composed of Males with 68% as opposed to females (32%). There is an opportunity to have a stronger customer base to attract more female shoppers in the market with marketing strategies designed for women (e.g. discounts during women's month).

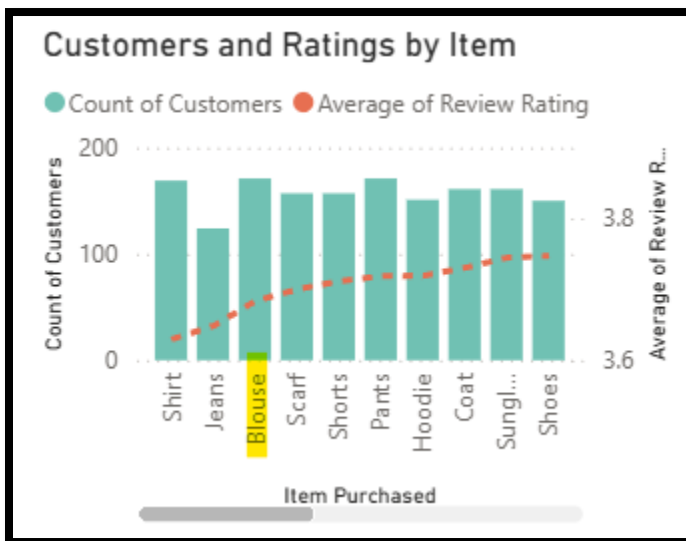
Products:

Clothing got the highest sales out of all item categories. It generated 44.73% of total sales of Blouse Barn (\$104,264). This is followed by Accessories (31.83%), Footwear (15.49%) and Outerwear (7.95%). It is further highlighted in Top 5 Most Purchased Items table where 4 out of 5 are from clothing category.



Top 5 Most Purchased Items	
Item Purchased	No. of Items Sold
Blouse	171
Jewelry	171
Pants	171
Shirt	169
Dress	166

However, despite performing best in terms of sales, this category performs the worst in terms of Review Ratings. One of the most frequently bought items (Blouse) is included in the worst reviews of all products in the shop.

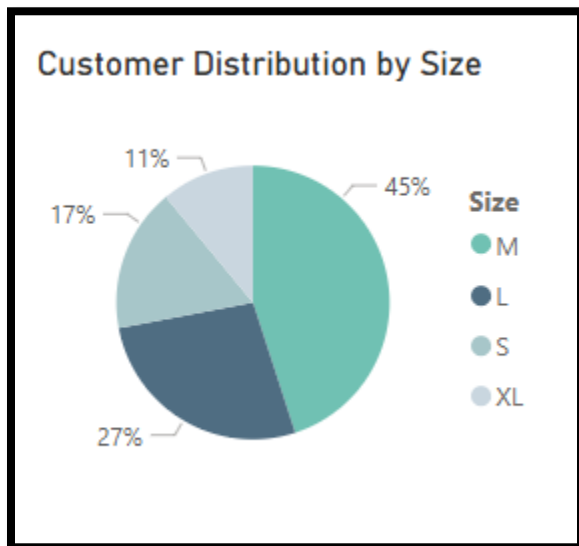


Top 5 Highest Rated Items	
Item Purchased	Avg. Review Rating
Gloves	3.86
Sandals	3.84
Boots	3.81
Hat	3.81
T-shirt	3.78

On the other hand, Accessories have the most interesting position. Second, to clothing, it generated \$74,200 in Sales with an average review rating of 3.77 (.53% higher than average).

Blouse Barn should take advantage of Accessories' ratings with its customers. Strategizing choices to increase its exposure, such as highlighting looks featuring best-reviewed accessories to its campaigns, supports its visibility to customer's perspective. Product bundling with other most purchased items such as dresses can help in increasing Accessories' volume in Sales while still retaining the familiarity with clothing's popularity.

This shift in focus will also be helpful in addressing the challenges with Clothing. Profit generated from increase sales of Accessories should be directed towards investigating and improving the quality of clothes to fix its bad reviews and strengthen customer's trust in the core revenue generating products in the shop.



Analysis of Customer's size preferences is also done to understand inventory movement. The analysis shows that Medium is the dominant choice, accounting for 45% of total purchases. Followed by Large (27%), Small (17%) and XL (11%). Blouse Barn should optimize its inventory by increasing stocks for Medium items and scale back from lower percentage sizes such as XL to maximize revenue and avoid over-inflating inventory carrying costs.

Season:

In terms of sales, there is no significant difference between seasons. Highest revenue is observed during Fall (\$60,018) and lowest during Summer (\$55,777). It shows good and stable adaptability of the shop's performance for each season.

On the other hand, review ratings contrast between the core customer's age groups (20-70) for different seasons. In *Fall*, customers aged 50-60 recorded the lowest average review rating with 3.63 (-3.2% than average). The analysis reveals that poorly rated items by the same age group include *Pants*, *Blouse* and *Jackets*. The *Jackets* ranked in the top 4 of most purchased items by these customers in Fall so these should be prioritized in Quality Assurance for this season.

In *Spring*, the second lowest review rating is recorded with 3.64 (-2.93% than average) by customers aged 40-50. Further analysis shows that these customers rated *Belt* (3.07) the lowest, but the item rated high in the overall ratings it received in Spring (3.78). It is beneficial for the shop to investigate the discrepancy of these ratings to understand the preferences that can accommodate this customer bracket better.

Review rating in *Summer* shows no significant drop. *Winter* observed changes in the usual high ratings of customers aged 20-30. This age group is averaging 3.79 across all seasons but recorded a -2.9% drop during *Winter* (3.68). Poorly rated items by them include *Backpacks*, *Handbags* and *Sandals*. *Backpacks* are the least purchased item in *Winter* but 16% of customers belong to this age group with a discrepancy of -19.2% in overall ratings of the item. It is recommended that Blouse Barn investigates feedback from these customers through promotional discounts, customer relations or review accessibility options (i.e. social media).

Different seasons also observed different prevailing products. It is understandable that items that fit the season—Jackets during Fall and Sunglasses during Summer—are the most purchased items in the shop. Using the frequency of item purchases graph is helpful in identifying product priorities in each season. Inventory should adjust according to these demands and observe the items needed to accommodate seasonal changes.

Conclusion:

David Rose's choices in the TV may have been unconventional, but we derived useful insights for Blouse Barn—should he have been an analyst in an alternate story.

The data shows that the core customer base of Blouse Barn are aged 20-70 years old with a demographic's majority consisting of *Men*. Low ratings are observed in customers 40-60 years old where implementing targeted feedback incentives is recommended to understand their preferences better. On the other hand, executing targeted marketing strategies designed for *Women* (*i.e discounts during Women's Month*) is recommended to drive a stronger customer base in Women's market.

In products, *Clothing* generated the most profit followed by *Accessories*. However, there is a noticeable difference between their review ratings with *Accessories* being received better by customers. Strategies in highlighting visibility in marketing campaigns and product bundling with popular products might help in increasing the volume of sales for *Accessories* while using additional profit to research and improve *Clothing's* quality to resolve its challenges. It is also noticed that *Medium* is the dominant choice in customer size preferences, accounting for 45% of total purchases. Followed by Large (27%), Small (17%) and XL (11%). Blouse Barn should optimize its inventory in favor of *Medium* sizes and scale back to lower demand sizes.

Seasons are seen having prominent changes in review ratings and purchased items. *Fall* recorded the lowest average review rating. It is from customers aged 50-60 with 3.63 (-3.2% than average) where *Jackets* are identified as a priority for quality checking in this season. *Spring* recorded second from the 40-50 aged group but discrepancy of ratings especially on *Belt* are observed with -18.73% than overall average it received in this season. Examining this discrepancy is beneficial for understanding their item sentiments. *Summer* showed no significant drop, but *Winter* recorded the lowest rating from a usually higher rating aged group (20-30). *Backpacks* are identified to be the lowest rated item from them with -19.2% in overall ratings of the item. It is recommended that Blouse Barn investigates feedback from these customers through promotional discounts, customer relations or review accessibility options (*i.e. social media*).

As expected, different seasons are seen with different purchase frequency where items that fit the season are bought more compared to others. Blouse Barn's inventory management should be able to adjust to different seasons using customer purchases data to accommodate demands and seasonal changes.